



9

TRANSFORMERS

9.1 Introduction

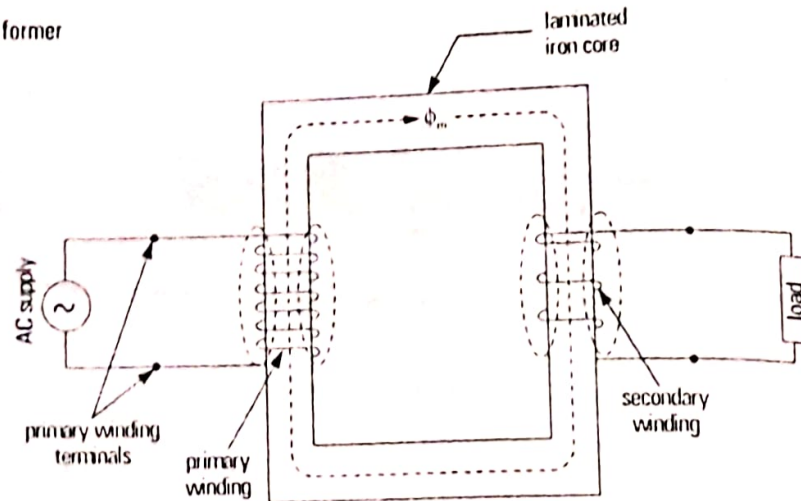
Transformers provide a means for converting an ac voltage from one value to another value. Electrical power is generated, distributed and utilised as ac because of transformers. Transformer is an important part of a transmission and distribution system. It is also used in high voltage and high current measuring instruments.

9.2 Principle of Operation

A transformer consists of two windings, primary winding and secondary winding wound on a laminated iron core as shown in Fig.9.1. When an alternating supply is connected across the primary winding, an alternating current flows through the primary winding. We have seen in chapter two that when a current flows through a coil a flux is produced inside the coil. Hence, when an alternating current flows through the primary winding an alternating flux ϕ is produced. This flux is linking the secondary winding also. We know that when the magnetic flux linking a coil changes an emf is induced in the coil. Hence, an emf is induced in the secondary winding. Thus, an alternating voltage connected to the primary terminal produces an alternating voltage across the secondary winding terminals.

FIG. 9.1

A simple transformer



In fact, the alternating flux induces a voltage in primary also. The magnitude of this emf, E_1 is given by Faraday's law:

$$E_1 = N_1 \frac{d\phi}{dt}$$

(N_1 is the number of turns on the primary winding) and the magnitude of the emf induced in the secondary is

$$E_2 = N_2 \frac{d\phi}{dt}$$

(N_2 is the number of turns on the secondary winding).

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

(9.1)

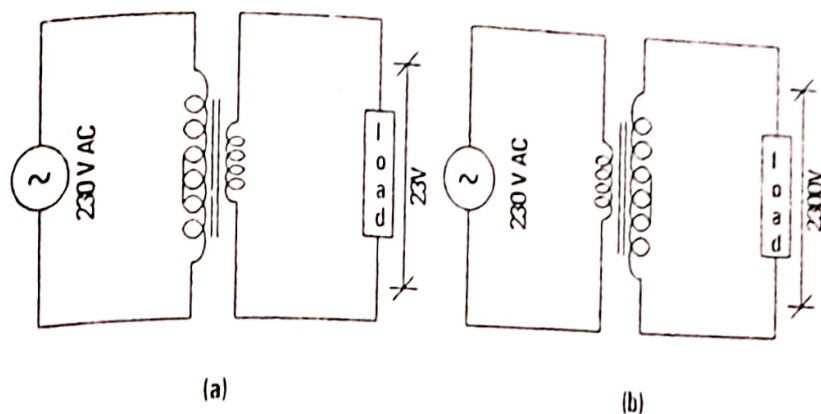
Therefore, if the number of turns on the secondary is less than that of the primary, the emf induced in the secondary will be less. For example, if $N_1 = 500$ and $N_2 = 50$, turns ratio of 10:1 is developed. Therefore, if 230 V ac is applied to the primary, a 23 V ac is developed across the secondary terminals. (Refer Fig.9.2(a)). Such a transformer is called a *step down transformer*.

If N_2 is greater than N_1 , E_2 will be greater than E_1 . Such a transformer is called a *step up transformer*. For example, if the primary winding has 50 turns and secondary winding has 500 turns, a turns ratio of 1 : 10 is developed. Therefore, if 230 V ac is applied to the primary, 10 times that voltage, or 2300 V ac, will be developed across the secondary terminals. (Refer Fig.9.2.(b))

In the last section, we have conveniently assumed that the flux linking the primary winding and the secondary winding is the same. This is not fully true.

FIG. 9.2

(a) Step down transformer
(b) Step up transformer



In an actual transformer, there will be some *leakage flux* as shown in Fig.9.3(a). Hence, flux linking the primary and secondary will not be the same. However, by appropriate construction, leakage flux could be reduced. In an ideal transformer, we assume there is no leakage flux. Transformer windings are made up of copper wire. There will be a small winding resistance for the primary winding and the secondary winding. However, in an ideal transformer, we assume the winding resistances to be zero.

In an actual transformer, when current flows through the windings, there will be I^2R loss (Cu loss). For an ideal transformer, since we have assumed $R = 0$, there will not be any copper losses. Moreover, we assume iron losses are also zero.

To summarise, for an ideal transformer, we assume

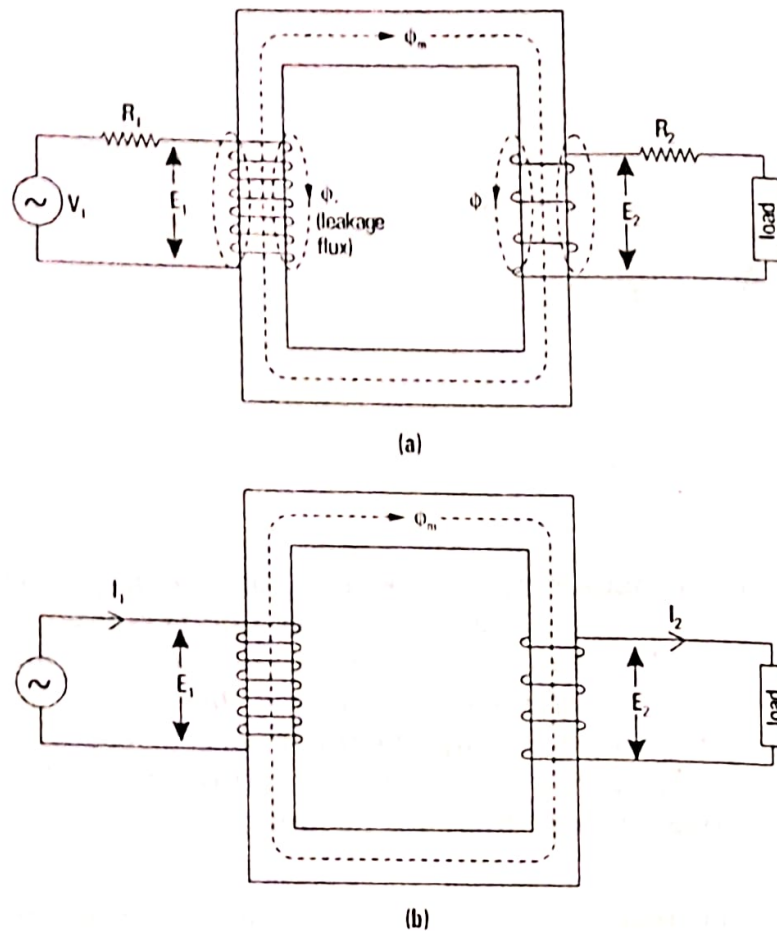
- i. there is no leakage flux,
- ii. winding resistances are equal to zero, and
- iii. no copper loss and iron loss

For most of the analysis, we assume the transformer to be ideal to make the analysis easy. In fact by proper construction, leakage flux could be reduced to almost zero; resistance of the winding is anyway negligible; and iron loss also could be minimized. Hence, the results obtained using the assumption of ideal transformer is more or less acceptable.

Suppose the load connected to the secondary of the transformer given in Fig.9.3(b) is resistive, then the power delivered by the (ideal) transformer is $E_2 I_2$. Since we have assumed losses to be zero, the power drawn from the supply $E_1 I_1$ should be equal to $E_2 I_2$.

FIG.9.3

(a) Actual transformer (b) Ideal transformer



That is

$$E_1 I_1 = E_2 I_2 \quad (9.1)$$

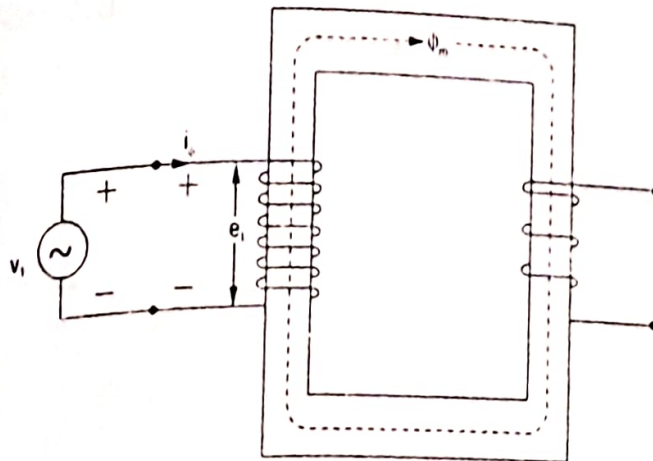
$$\frac{E_1}{E_2} = \frac{I_2}{I_1} \quad (9.2)$$

Using (9.1), we get an important equation for the transformer.

$$\frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} \quad (9.3)$$

9.3 EMF Equation

Consider the transformer on no load given in Fig.9.4. When alternating voltage, $v_1(t) = V_{\max} \cos \omega t$, is applied, an alternating current known as excitation current, i_0 , flows through the coil. This excitation current will be lagging the voltage by 90° . (Recall current in an inductive circuit lags the voltage by 90°). Therefore, the excitation current $i_0(t)$ equal to $I_{\max} \cos(\omega t - 90) = I_{\max} \sin \omega t$.



This will in turn produce an alternating flux $\phi = \phi_{\max} \sin \omega t$. We know, when the flux linking a coil changes an emf is induced. The instantaneous value of this voltage* is given by Faradays law.

$$e_1(t) = -N_1 \frac{d\phi}{dt} = -N_1 \frac{d}{dt} \phi_{\max} \sin \omega t$$

$$= -N_1 \phi_{\max} \omega \cos \omega t = -E_{\max} \cos \omega t$$

here, $E_{\max} = N_1 \omega \phi_{\max}$ (but $\omega = 2\pi f$)

$$= N_1 \times 2\pi f \phi_{\max}$$

$$E_{\text{rms}} = \frac{E_{\max}}{\sqrt{2}} = \frac{2\pi f N_1 \phi_{\max}}{\sqrt{2}} = 4.44 \phi_{\max} f N_1$$

$$E = 4.44 \phi_{\max} f N_1$$

The above equation is called the emf equation of the transformer.

4 Constructional Details

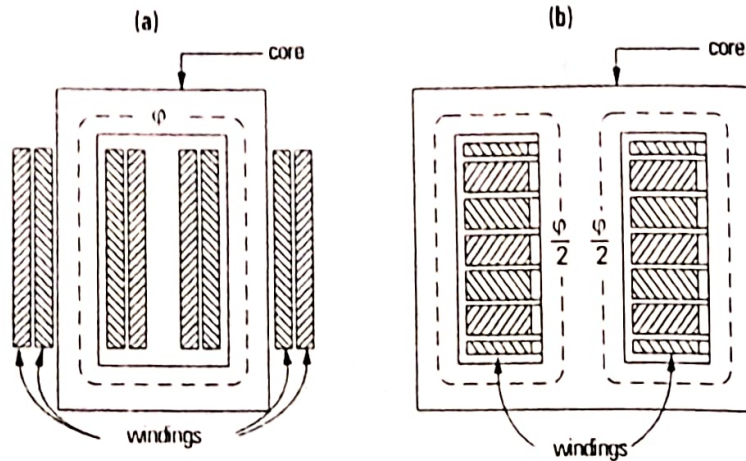
The two main parts of a transformer are the iron core and the windings. To minimize eddy current losses, transformer core is made of thin silicon steel laminations insulated from each other using thin coat of varnish. There are two types of construction: core type and shell type. In the core type transformer (Fig.9.5(a)), windings are wound around two legs of a rectangular magnetic core as shown in Fig.9.5a. In this type of construction, same flux flows through the entire core. Hence, area of cross section of the core is same for the entire core.

*The direction of this emf, according to Lenz's law, is in a direction to oppose the cause which produces it. That is, e_1 opposes v_1 as shown in Fig.9.5.

In shell type transformer, the windings are wound around the central leg of a three-legged core as shown in Fig.9.5(b). Since the side limbs carries only half the flux, the area of cross section of the side limbs is half of the area of cross section of the central limb.

FIG.9.5

(a) Core type (b) Shell-type transformer



In the simple transformer shown in Fig.9.1, we have shown primary windings and secondary windings on separate limbs. However, this arrangement is not efficient as the two windings are well separated from one another, resulting in large magnetic leakage. In practical transformers, windings are placed as close as possible. This could be achieved by *concentric construction* or *sandwich construction*. In the concentric connection of winding, primary winding and secondary windings are wound concentrically around the iron core. In the sandwich construction of windings, primary winding is sandwiched between two sets of secondary windings.

Ventillation spaces are left in the windings to allow space for coolant to circulate, which may be either air or oil. In large power transformers, cooling is a major issue, and mostly oil is used as the coolant.

9.5 Transformer Losses and Efficiency

There are two types of losses in a transformer: (i) copper loss (ii) iron loss. Cu loss is the loss occurring in the copper conductors of the winding. When a current of I_1 A flows through the primary winding, $I_1^2 R_1$ watt is dissipated as heat (where R_1 is the resistance of the primary winding). Similarly $I_2^2 R_2$ watt is dissipated as heat in the secondary winding. (Where I_2 and R_2 are the current and resistance of the secondary winding). Cu loss are also known as $I^2 R$ losses, and as seen from the expression is depended on the current, or load on the transformer.

Iron loss consists of two components: eddy current loss and hysteresis loss. Parts of the iron core can act as closed conductors. Since the magnetic flux in the iron core varies (or pulsates), an emf is induced in these closed circuit which causes a circulating current to flow in the iron core ('conductor'). Such a current in turn will cause a power loss (an I^2R loss) known as eddy current loss.

To summarise, Cu loss is due to the heat dissipation (I^2R) in the primary and secondary winding when load current flows through it. It depends on the load current. Iron loss (eddy current loss + hysteresis loss) is because of the alternating flux in the iron core. Iron loss is more or less constant. It can be reduced by using thin laminated silicon steel for core construction.

Efficiency of a transformer depends on the load current as losses (Cu loss) are dependent on load. Hence, the efficiency of a transformer is defined for a particular load current. The efficiency of a transformer for a particular electrical load is given by

$$\begin{aligned}\eta &= \frac{\text{output power}}{\text{output power} + \text{losses}} \\ &= \frac{\text{output power}}{\text{output power} + Cu \text{ loss} + \text{Iron loss}}\end{aligned}$$

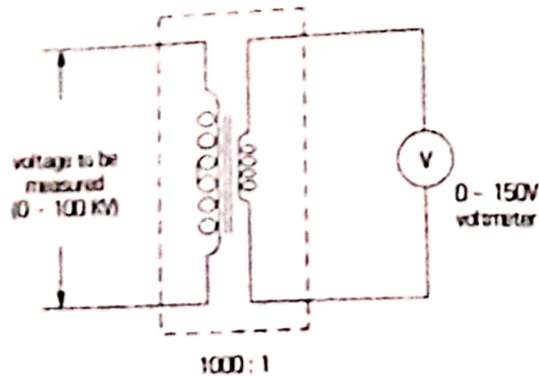
A well designed transformer has relatively low power loss (compared to the output of the transformer), hence the efficiency is very high (above 90%).

9.6 Instrument Transformers : CT and PT

Instrument transformers are used as an aid to measure high voltages and high currents. It is impractical to design an ammeter which can operate at thousands of ampere. Hence, to measure high voltages and high currents these quantities are transformed to a smaller level, and conventional instruments are used. The transformer used to reduce the voltage (potential) level to a manageable value is called a potential transformer (PT). The transformer used to reduce the current level is called a current transformer or CT.

PT's are step-down transformers. For example, to measure voltages of the order of 100 KV, a step down transformer of turns ratio 1000:1 is used. For example, if the voltage across the primary is 90 KV, the voltage across the secondary will be only 90 V. If the turns ratio is $N:1$, the actual voltage can be calculated by multiplying the voltmeter reading by N .

FIG. 9.6
Potential transformer



Current transformer's (CT's) are step up transformers. One main advantage of CT is the fact that to measure current through a circuit (say overhead line) we need not open the circuit. Usually the overhead line (or path in which the current is to be measured) acts as a single turn primary winding. Secondary winding consists of hundreds of turns so that the secondary current is between 0 and 5 A.

To calculate the number of turns, recall the important relationship of a transformer:

$$\frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

For example, to measure current upto 1000 A, we require the secondary current to be less than 5 A.

i.e., when $I_1 = 1000$ A, we want $I_2 = 5$ A

$$\therefore \frac{N_1}{N_2} = \frac{5}{1000}$$

Since $N_1 = 1$,

$$N_2 = 200$$

That is, to measure current upto 1000 A use a CT of turns ratio 1 : 200.

■ SOLVED PROBLEMS

9.1 An ideal single phase transformer has 500 turns on the primary winding, and 100 turns on the secondary winding. If the supply voltage is 1000 V, find the turns ratio and the secondary voltage. If the secondary load is a noninductive resistor of 10 Ω resistance, calculate (i) current through the secondary winding (ii) current through the primary winding and (iii) power drawn by the load.

Solution

Given

$$N_1 = 500; N_2 = 100; E_1 = 1000 \text{ V}; R_L = 10 \Omega$$

Turns ratio ?

$$E_2 = ?; I_2 = ?; I_1 = ?; P_2 = ?$$

$$\text{Turns ratio} = \frac{N_1}{N_2} = \frac{500}{100} = 5:1$$

We know,

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

$$E_2 = 1000 \times \frac{100}{500} = 200 \text{ V}$$

$$I_2 = \frac{E_2}{R_L} = 20 \text{ A}$$

$$\frac{E_1}{E_2} = \frac{I_2}{I_1}$$

$$\frac{1000}{200} = \frac{20}{I_1}$$

$$I_1 = 4 \text{ A}$$

$$\text{Power drawn by the load} = E_2 I_2 = 200 \times 20 = 4000 \text{ W}$$

9.2 A 6600/550 V transformer has an iron loss of 350 W and a full-load copper loss of 415 W. If the full load secondary current is 45 A at 0.6 lagging pf, calculate the full load efficiency of the transformer.

Solution

Given

$$E_1 = 6600 \text{ V}; E_2 = 550 \text{ V}$$

$$\text{Iron loss} = 350 \text{ W}$$

$$\text{Full load Copper loss} = 415 \text{ W}$$

$$\text{Full load secondary current } I_2 = 45 \text{ A}$$

$$\text{Power factor } (\cos \phi) = 0.6$$

$$\eta = \frac{\text{output power}}{\text{output power} + \text{losses}}$$

$$\text{Output} = E_2 I_2 \cos \phi = 550 \times 45 \times 0.6 = 14850 \text{ W}$$

$$\text{Losses at Full load} = 350 + 415 = 765 \text{ W}$$

$$\eta = \frac{14850}{14850 + 765} = 95.1\%$$

9.3 The iron loss of 230/115 V 5 KVA transformer is 200 W. The copper loss at full load is 250 W. Find efficiency of the transformer when delivering: (i) full load power at u.p.f. (ii) half full load power at u.p.f. and (iii) half full load power at 0.8pf lagging.

Solution

Note : Usually transformer ratings are given in KVA. KVA rating is the product of secondary voltage and secondary current.

$$\text{KVA rating} = E_2 I_2$$

Given

$$E_1 = 230 \text{ V}; E_2 = 115 \text{ V}$$

$$\text{Rating } (E_2 I_2) = 5 \text{ KVA} = 5000 \text{ VA}$$

$$\text{Cu loss at full load} = 250 \text{ W}$$

$$\text{Iron loss} = 200 \text{ W}$$

(i) η_{FL} at u.p.f

$$\eta_{FL} = \frac{\text{output}}{\text{output} + \text{F.L.Cu loss} + \text{Iron loss}}$$

$$\text{output} = E_2 I_2 \cos \phi = 5000 \times 1 = 5000 \text{ W}$$

$$\eta_{FL} = \frac{5000}{5000 + 250 + 200} = 91.75\%$$

(ii) η at Half Load at u.p.f. (η_{HL} (u.p.f.))

To calculate η_{HL} , we have to calculate the Cu loss at Half load. We know, Cu loss is proportional to square of the current. Hence, at Half load, Cu loss = $\frac{1}{4}$ Cu loss at full load. In general, Cu loss at a fraction of full load = $x^2 \times$ Cu loss at full load

$$\text{Cu loss at } x \times \text{FL} = x^2 \times \text{FL Cu loss}$$

$$\begin{aligned} \therefore \text{Cu loss at } 0.5 \times \text{FL} &= (0.5)^2 \times \text{FL Cu loss} \\ &= 0.5^2 \times 250 = 62.5 \text{ W} \end{aligned}$$

$$\text{Output} = 0.5 \times \text{KVA rating} \times \text{p.f.} = 0.5 \times 5000 \times 1 = 2500 \text{ W}$$

$$\eta_{HL} (\text{u.p.f.}) = \frac{2500}{2500 + 62.5 + 200} = 90.50\%$$

η at half full load at 0.8 p.f. lagging ($\eta_{FL}(0.8pf)$)

$$\text{Output} = 0.5 \times 5000 \times 0.8 = 2000 \text{ W}$$

$$\text{Cu loss at Half load} = (0.5)^2 \times 250 = 62.5$$

$$\eta_{HL}(0.8pf) = \frac{2000}{2000 + 62.5 + 200} = 88.39\%$$

Note : A general formula for finding efficiency of a transformer at any load and power factor is given by

η at $x \times FL$

$$= \frac{x \times \text{KVA rating} \times \text{power factor}}{x \times \text{KVA} \times \text{Power factor} + x^2 \times \text{FL Cu loss} + \text{Iron loss}}$$

Draw the wave forms of applied voltage, magnetisation current and flux in an ideal transformer.

ation

the applied voltage be,

$$v(t) = V_{\max} \cos \omega t$$

know the magnetisation current ($i_o(t)$) is lagging the voltage by 90° .

$$I_o(t) = I_{\max} \cos(\omega t - \pi/2) = I_{\max} \sin \omega t$$

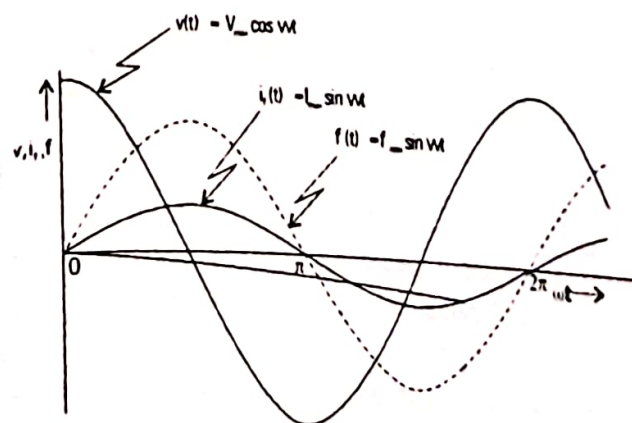
flux produced by the current is in phase with the magnetisation current i.e.,

$$\phi(t) = \phi_{\max} \sin \omega t$$

The wave forms of $v(t)$, $i_o(t)$, $\phi(t)$ is given in Fig.9.7.

1.7

tion of problem 9.4

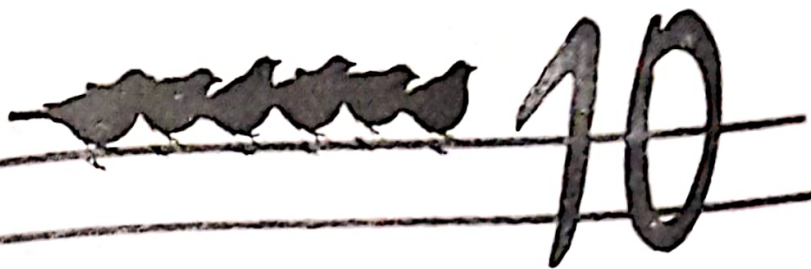


REVIEW QUESTIONS**Short Answer Type**

1. What is meant by the 'coupling coefficient' of a magnetic circuit?
2. A transformer has 100 turns of wire on its primary winding and 20 turns of wire on its secondary winding. If the supply voltage is 200V, calculate the secondary voltage.
3. How can eddy current losses be reduced ?
4. Why is copper loss dependent on load current and core loss dependent on applied voltage ?
5. Does the efficiency of a transformer depend on load? Why?
6. State the equation for calculating the efficiency of a transformer at a specified load and power factor.
7. If the Cu loss at rated load is P_{cu} and iron loss at rated load is P_{iron} . Find the load corresponding to maximum efficiency in term of P_{cu} and P_{iron} .
8. "The efficiency of transformers depends on the p.f.". Why?
9. Explain how potential transformer can be used to extend the range of a voltmeter.
10. What are the advantages of using CT for measuring high currents ?
11. How is a CT different from a step up distribution transformer?
12. Explain the principle of operation of a transformer.
13. Explain the difference between an ideal transformer and an actual transformer.
14. What are the various losses in a transformer? How can they be minimised?
15. Why transformer ratings are expressed in AVA ?

PROBLEMS

1. The full load copper loss and iron loss of 230/460 V 10 KVA transformer are 200 W and 220 W respectively. Find the efficiency of the transformer when delivering (i) full load at u.p.f., (ii) half full load at 0.8 p.f. lagging, (iii) half full load at 0.6 p.f. lagging and (iv) 70% of full load at 0.8 p.f. lagging.



DC GENERATORS

10.1 Introduction

DC generators are used to generate dc power. Even though the present day power distribution is predominantly ac, there are situations where dc power is required. For example, the field winding of an ac generator (alternator) requires dc supply. DC generator operation is based on Faraday's law of electromagnetic induction, which says that whenever there is a relative motion between a conductor and a magnetic field an emf is induced. In this chapter we learn the principle of operation, constructional details and types of dc generator.

10.2 Principle of Operation

The working of a d.c. generator is based on the principle of electromagnetic induction which you have already studied. According to this principle, when a conductor moves through a magnetic field an emf is induced.

Consider the conductor AB placed in the magnetic field as shown in Fig.10.1. We know that magnetic lines of force travel from the north pole to the south pole. When the conductor is moved in the downward direction an emf is induced, and since there is closed path through the meter an induced current will flow through the conductor. When the conductor is moved in the upward direction, the emf induced is in the opposite direction.